

Convergence of the Multi-Period Binomial Model to the Black-Scholes Framework

Purpose

This report examines how the Black-Scholes model arises as the continuous-time limit of a multi-period binomial tree. We begin by providing background on derivative markets and option pricing theory, followed by the construction of both the single-period and multi-period binomial models. We include a detailed overview of the Black-Scholes framework and its corresponding assumptions.

We then begin the mathematical portion of our report by showing that the multi-period binomial pricing model can be represented by the Cox-Ross-Rubinstein model. We expand on the CRR framework by applying the Central Limit Theorem and appropriate Taylor expansions, we show that the binomial distribution tends to a normal distribution as N becomes large. Together these components establish the convergence of the binomial option pricing model as described in CRR to the Black-Scholes model as $N \rightarrow \infty$.

Numerical simulations from our C++ implementation illustrate the convergence of the multi-period binomial model to the Black-Scholes Framework. The simulations show interesting observations as we plot the convergences over a range of hyperparameter values (S, K, T, r, σ). We discuss the appropriate uses of both the black-scholes and binomial models to evaluate and approximate the price of American options.

The analysis is developed within a no-arbitrage and risk-neutral valuation framework consistent with the material taught in MTH771P and MTH776P at Queen Mary University of London.

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1 Introduction to Derivative Markets

A financial derivative is a financial instrument whose value depends on the price of some underlying asset such as a stock, index, interest rate, currency or commodity.

1.1 Introduction to Options

A financial option contract is a derivative contract which gives the buyer the right, or option to purchase or sell an asset at a fixed price at some future date, which we will call time T . The agreed upon price is referred to as the strike price, and is denoted by K , and the future date is called the expiration date, and is denoted by T . A call option is an option which gives the owner the right to purchase the asset at time T , whereas a put option gives the owner the right to sell an asset at time T .

1.1.1 European and American Options

Options can be classified as either *American* or *European*. American options give the owner the right to exercise the option at any time of ownership before time T , while European Options can only be exercised at expiration. In this report we will denote European call and puts by C, P and American call and puts by C^A, P^A , and it follows that $C_t \leq C_t^A, P_t \leq P_t^A, \forall t$. In this report unless explicitly mentioned we will be speaking about European Options.

1.2 Obligation of the Option's Writer

Thus far we've highlighted options from the perspective of the buyer. The option "seller" or "writer" is the counterparty who

sells the option, receives an upfront premium and takes the obligation associated with the contract. If the buyer chooses to exercise the option the writer of a call option is obligated to sell or deliver the underlying asset to the buyer at price K , whereas the writer of a put is obligated to buy the underlying from the buyer at price equal to K .

1.3 Value of an option

The value of an option consists of two parts:

- **Intrinsic value:** the immediate exercise value, $\max(S_t - K, 0)$ for calls and $\max(K - S_t, 0)$ for puts.
- **Time value:** the additional value arising from the uncertainty in future price movements.

1.4 Moneyness of an option

An option's *moneyness* describes the relationship between the underlying and strike prices. An option is either considered in-the-money, at-the-money, or out-of-the money at all points in its life. An option's moneyness depends on the price of an asset relative to the strike price K .

- **At-the-money (ATM):** $S_t = K$.
- **In-the-money (ITM):** call: $S_t > K$, put: $S_t < K$. An in-the-money the option has positive intrinsic value.
- **Out-of-the-money (OTM):** call: $S_t < K$, put: $S_t > K$.

1.5 Payoff of Options

The payoff of an option is the amount received by the holder at expiry T if the option is exercised. Since the holder has a right but not an obligation to exercise, the payoff is always non-negative.

A European call option is exercised only if the asset price at expiry exceeds the strike price. Conversely, a European put option is only exercised if the asset price is less than the strike price at expiry. At time T , the value of the call and put are

$$C_T = \max(S_T - K, 0), \quad P_T = \max(K - S_T, 0).$$

1.6 Option Strategies

Options are used in financial markets for three primary purposes: hedging, speculating, and arbitraging.

Hedging is a risk management strategy in which market participants use options to reduce or offset the risk of adverse price movements in an underlying asset. It is common for market participants to hedge their long/short equity positions with either puts or calls to reduce their downside exposure.

Speculation refers to taking a view on future price movements or volatility. Options allow traders to express directional or volatility-based views with leverage and limited downside risk.

Arbitrage involves constructing portfolios that exploit mispricings between related assets. In its simplest form, this may involve a violation of put-call parity, while in more complex settings traders may use multi-leg derivative strategies to take advantage of subtle inconsistencies across strikes or maturities. The idea of arbitrage expressed here shows why we price models under the assumption of no-arbitrage, which will be further explained within section 2.5. Firms and market makers often seek to take advantage of arbitraging using proprietary pricing models.

1.7 Importance of Option Pricing in Financial Markets

Option pricing plays an important role in modern financial markets. Because many assets contain embedded optionality and because derivatives are traded across a wide range of markets, a consistent and theoretically sound pricing framework is essential.

1.7.1 Trading and Market Making

Options provide a common language for valuing and comparing payoffs across strikes, maturities, and underlying assets. Market makers rely on option pricing models to quote fair bid-ask prices and to hedge the risks associated with holding large inventories of options. Consistency in valuation across contracts prevents arbitrage and supports market liquidity.

For active traders, options serve as precise tools for expressing an investment view. Traders can construct higher complexity positions using options. Examples are the ability to take targeted positions on volatility, create spread trades to exploit relative pricing distortions, or set up structured trades wherein risk is capped but potential gain is not limited. In these markets, European options dominate because their simplicity promotes liquidity and they feature only a single exercise date.

1.7.2 Market Expectations and Volatility

Stock options express market sentiment about future volatility. The shape and form in which options prices take on shape

represents the market expectations for the movements in underlying assets. These patterns, as well as the implied volatility surface, allow volatility analysts to design the implied volatility landscape that predicts investors' stock portfolio behaviour. European options play a main role in analysis because they are fixed and correspond to follow agreements timeframe.

1.7.3 Structured Products and Fixed Income

Many financial instruments contain embedded options, such as callable bonds, convertibles, and structured notes. Option pricing can be used to break these instruments down to evaluate their intrinsic values. Early repayment/conversion in the fixed income and credit market is essentially an option exercise. The American-style models capture these features correctly by linking bond valuation models to credit risk and derivative valuation in one unified theoretical construct.

1.7.4 Regulation, Accounting, and Model Validation

Regulators and financial institutions rely on model-based valuation when market prices are unavailable. Fair-value accounting standards, capital requirements, and risk management frameworks often depend on option pricing models to assess exposures. Because model errors can lead to mispricing and misallocation of risk, institutions must validate and stress test their models, highlighting the importance of mathematically robust pricing methods.

1.8 Option Greeks

The option Greeks measure the sensitivity of the price of the option to change in the underlying parameters. Each Greek corresponds to a partial derivative of the option price with respect to a parameter. The following are the most commonly referred to greeks.

Delta (Δ): $\Delta = \frac{\partial V}{\partial S}$ measures the change in the option value V for a one unit price change in the underlying asset price S .

Gamma (Γ): $\Gamma = \frac{\partial^2 V}{\partial S^2}$ measures the change in Delta for a one unit change in the price of the underlying asset price. It captures the curvature of the option value with respect to S .

Vega: $\text{Vega} = \frac{\partial V}{\partial \sigma}$ measures the change in the option value for a 1% change in the volatility σ .

Theta (Θ): $\Theta = \frac{\partial V}{\partial t}$ measures the change in the option value in time to maturity. Theta reflects the time-decay of the option's value, and is usually quoted on a per-day basis.

Rho (ρ): $\rho = \frac{\partial V}{\partial r}$ measures the change in the option value for a 1% change in the risk-free interest rate r .

2 Discrete Option Pricing Models

Option pricing is a fundamental topic in financial mathematics, with significant implications for risk management, portfolio optimization, and derivative valuation. Accurate pricing models help investors, traders, and institutions make informed decisions, manage risk exposures, and identify arbitrage opportunities. Two primary approaches exist for modelling option prices: discrete-time and continuous-time frameworks.

The binomial model operates in discrete time, dividing an option's life into several intervals and modelling the underlying asset's potential movements step by step. A multi-period binomial tree allows for detailed modelling of future price paths, early exercise opportunities, and dividend

adjustments.

In contrast, continuous-time models provide analytical solutions under stronger assumptions about market dynamics. The multi-period binomial model serves as a discrete-time framework capable of approximating the behaviour of continuous-time models as the number of steps increases.

2.1 Single-Period Binomial Model

The binomial method assumes that the underlying asset price can move to only two possible states over a given time interval. This simple two-state structure captures the core uncertainty of financial markets while remaining mathematically tractable. It forms the foundation for modelling asset prices, valuing derivatives, and assessing risk in discrete time.

The single period binomial model only considers $t = 0$ and $t = T$. Over this one-period horizon, the price of a risky asset may move up or down. This basic structure allows us to analyse option pricing, hedging, and no-arbitrage in a clear and intuitive setting.

At time 0 the stock price is S_0 . In the single-period binomial model, after one period the stock price S_1 can take one of two possible values:

$$S_1 = \begin{cases} uS_0, & (\text{up state}), \\ dS_0, & (\text{down state}), \end{cases}$$

where $u > 1$ and $0 < d < 1$. A risk-free asset (such as a bank deposit or treasury bill) grows from 1 to $1 + R$ over the same period.

Real-world probabilities of an upward movement p and a downward movement $1 - p$ represent true market beliefs, but these are not used directly for pricing derivatives. Instead, pricing is based on risk-neutral probabilities, which arise from the no-arbitrage condition.

2.2 Option Pricing in the Single-Period Model

For a European call option with strike price K , the payoff at maturity depends on the stock price movement:

$$C_u = \max(0, uS_0 - K), \quad C_d = \max(0, dS_0 - K).$$

Instead of attempting to predict the option's value, we construct a replicating portfolio that matches the option payoff in both states. By the *Law of One Price*, two portfolios with identical payoffs in all future states must have the same present value.

2.3 Replication and Hedging Arguments

Consider a portfolio consisting of:

- Δ shares of the stock,
- β units of cash in the bank.

Portfolio values:

$$V_0 = \Delta S_0 + \beta,$$

$$V_u = \Delta S_1^{(u)} + \beta(1 + R), \quad V_d = \Delta S_1^{(d)} + \beta(1 + R).$$

Replication requires:

$$V_u = C_u, \quad V_d = C_d.$$

Solving:

$$\Delta = \frac{C_u - C_d}{S_1^{(u)} - S_1^{(d)}},$$

$$\beta = \frac{C_d S_1^{(u)} - C_u S_1^{(d)}}{(S_1^{(u)} - S_1^{(d)})(1 + R)}.$$

Thus, the option price is:

$$V_0 = \Delta S_0 + \beta.$$

2.3.1 Delta Hedging

The replication concept can be interpreted as a hedging strategy. Consider a portfolio with:

- one long option,
- short Δ shares of the stock.

At time 0:

$$\Pi_0 = V_0 - \Delta S_0.$$

At time 1:

$$\Pi_1^{(u)} = C_u - \Delta S_1^{(u)}, \quad \Pi_1^{(d)} = C_d - \Delta S_1^{(d)}.$$

Using the same Δ from replication, the hedged portfolio becomes risk-free:

$$\Pi_1^{(u)} = \Pi_1^{(d)} = \Pi_1.$$

Its present value is the discounted value of the risk-free payoff:

$$V_0 = \Delta S_0 + \frac{\Pi_1}{1 + R}.$$

2.4 Arbitrage and No-Arbitrage Conditions

An arbitrage opportunity is a trading strategy with the following properties:

1. It requires no initial investment.
2. Its payoff is never negative at maturity.
3. It has a strictly positive probability of yielding a profit.

The existence of arbitrage would imply that the market offers a risk-free gain, or "free money." Although arbitrage situations may arise temporarily in real markets, efficient markets ensure that they are corrected quickly. For practical purposes, and throughout this report, we assume the *absence of arbitrage*.

2.5 Risk-Neutral Probabilities

We will now introduce the idea of the risk-neutral probability measure $\mathbb{Q}$. The risk-neutral probability measure $\mathbb{Q}$ is a probability measure under which the expected one-period growth of the asset price is equivalent to the single period risk-free rate R . That is,

$$S_0 = \frac{1}{1 + R} \mathbb{E}^{\mathbb{Q}}[S_1]$$

For the sake of the binomial model we consider the risk-neutral probabilities of an "up" and "down" move, which we denote q_u and q_d . Note that q_u and q_d are not real-life observed probabilities, rather the probabilities associated with the binomial model in a risk-neutral environment. We define q_u and q_d as follows:

$$q_u = \frac{S_0(1 + R) - S_1^{(d)}}{S_1^{(u)} - S_1^{(d)}}, \quad q_d = 1 - q_u.$$

Going forward throughout this report unless otherwise specified, we will be pricing options under the risk neutral measure $\mathbb{Q}$ as defined here.

2.6 Applying the Risk-Neutral Measure

Now that we've derived and set-up a risk-neutral environment for the binomial model, we are able to price the option under the risk neutral measure. At time $t = 1$, the asset will have price at $S_1^{(u)}$ with R.N. probability q_u , or it will have the price $S_1^{(d)}$ with R.N. probability q_d . We write the price of the option at time 0 as:

$$V_0 = \frac{1}{1+R} [q_u V_1^{(u)} + q_d V_1^{(d)}].$$

2.7 Fundamental Theorems of Asset Prices

2.7.1 First Fundamental Theorem of Asset Pricing

The first fundamental theorem of asset pricing states that a model is free of arbitrage if and only if a risk-neutral measure exists. This supports our previous argument to price options using the assumptions of no-arbitrage.

2.7.2 Second Fundamental Theorem of Asset Pricing

The second fundamental theorem of asset pricing states that in a market that is free of arbitrage, the market is complete if and only if the risk-neutral measure is unique. The importance of the theorem with respect to our research is that it guarantees the uniqueness of derivative prices.

3 The Black-Scholes Model

3.1 Overview

The Black-Scholes model was first introduced by Black and Scholes in 1973. Black and Scholes derived the model by considering the possibility of continuously replicating the payoff of European option using the stock and a risk-free asset. Note that the dynamic replication leads to the Black-Scholes PDE. Recall by law of one price, the option must therefore have the same value as its replicating portfolio. The Black-Scholes model can be viewed as the continuous-time limit of the multi-period binomial model introduced earlier.

3.2 Model Assumptions

The Black-Scholes framework relies on the following assumptions:

1. The stock price follows the following differential equation:

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where μ is the drift term, σ is the volatility, and W_t is a standard Brownian motion.

2. The risk-free interest rate r and volatility σ are constant.
3. There are no transaction costs or taxes.
4. Trading is continuous.
5. There are no arbitrage opportunities.
6. The stock pays no dividends (Merton's extension allows for continuous dividends).

3.3 Distribution of the Underlying Asset

The stock price follows a geometric brownian process as follows:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

The PDE has the explicit solution

$$S_T = S_0 \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) T + \sigma W_T \right).$$

Hence,

$$\ln \left(\frac{S_T}{S_0} \right) \sim \mathcal{N} \left(\left(\mu - \frac{1}{2} \sigma^2 \right) T, \sigma^2 T \right).$$

Under the risk-neutral measure $\mathbb{Q}$, the drift becomes r , giving

$$\log \left(\frac{S_T}{S_0} \right) \sim \mathcal{N} \left(\left(r - \frac{1}{2} \sigma^2 \right) T, \sigma^2 T \right).$$

That is, for a small Δt we can see that the stock price is log-normally distributed with mean $r - \frac{1}{2} \sigma^2 \Delta t$ and variance $\sigma^2 \Delta t$.

$$\log \left(\frac{S_{t+\Delta t}}{S_t} \right) \sim \mathcal{N} \left(\left(r - \frac{1}{2} \sigma^2 \right) \Delta t, \sigma^2 \Delta t \right).$$

3.4 The Black-Scholes Equation

3.4.1 Closed-Form Solution

The price of a European call option is

$$C_0 = S_0 N(d_1) - K e^{-rT} N(d_2),$$

where

$$d_1 = \frac{\log(S_0/K) + (r + \frac{1}{2} \sigma^2)T}{\sigma \sqrt{T}}, \quad d_2 = d_1 - \sigma \sqrt{T},$$

and $N(\cdot)$ denotes the standard normal cumulative distribution function.

The European put price follows from put-call parity:

$$P_0 = K e^{-rT} N(-d_2) - S_0 N(-d_1).$$

3.4.2 Interpretation of $N(d_1)$, $N(d_2)$

Taking the derivative of the price of the option gives

$$\frac{\partial C_0}{\partial S_0} = N(d_1) \quad \frac{\partial P_0}{\partial S_0} = -N(-d_1)$$

so the $N(d_1)$, $-N(-d_1) = N(d_1) - 1$ terms are the delta for the call and the put respectively.

The term $N(d_2)$ represents the probability that the price of the stock is greater than the strike, while $N(-d_2)$ represents the probability that the strike price will be greater than the price of the stock. Note that the above results only hold under the risk neutral measure $\mathbb{Q}$.

3.5 Merton's Extension for Dividend-Paying Assets

In 1973, Robert Merton generalised the Black-Scholes framework to incorporate European options on dividend-paying assets in his article "*Theory of Rational Option Pricing*". Merton extended the original framework by allowing the underlying asset to pay dividends at a continuous proportional rate q (Merton 1973). The stock price component of the Black-Scholes equation is changed from S_0 to $S_0 e^{-qT}$, reflecting the price of the asset discounted for the dividends paid along time T . The Black-Scholes-Merton equations for a dividend-paying asset given by:

$$C_0 = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2),$$

$$P_0 = K e^{-rT} N(-d_2) - S_0 e^{-qT} N(-d_1).$$

3.6 Model Parameters

The Black-Scholes formula depends on the following parameters:

- S_0 : current price of the underlying asset,
- K : strike price of the option,
- T : time to maturity,
- r : continuously compounded risk-free interest rate,
- σ : volatility of the underlying asset.

Volatility typically has the largest impact on the option price. In practice, the volatility parameter is inferred from market prices and is known as the *implied volatility*.

3.6.1 Implied Volatility

The Black-Scholes formula requires the volatility parameter σ as an input, but volatility is not directly observable in markets. While volatility is an input of the Black-Scholes equation, market participants invert the Black-Scholes equation to determine the value of σ that matches market option price.

Implied volatility reflects the market's assessment of future uncertainty. Given that volatility remains an input in options pricing models, market participants often use complex models to approximate volatility.

3.7 The Option Greeks

Now that we have derived the Black-Scholes pricing formula, we return to our earlier definitions of the option Greeks. Since each Greek corresponds to a partial derivative of the option value with respect to a model parameter, we can apply these definitions directly to the Black-Scholes formula, which gives explicit expressions for the values of the greeks and consequently sensitivities of European option prices to changes in the underlying parameters.

4 Convergence of the Multiperiod Binomial Model

4.1 Multi-Period Binomial Option Pricing Model

The multi-period binomial model divides the option's life into $N > 1$ discrete intervals, forming a lattice of possible underlying asset prices. At each step, the asset may move up or down, and option values are computed recursively from terminal nodes using risk-neutral probabilities.

Key assumptions include:

- the underlying asset follows a multiplicative binomial process,
- markets are frictionless,
- trading occurs at discrete intervals,
- early exercise is possible for American options,
- risk-neutral probabilities determine expected discounted payoffs.

The multi-period structure improves upon the single-period model by increasing granularity, capturing early exercise decisions more effectively, and reducing approximation error. Terminal payoffs are calculated at final nodes, and backward induction determines earlier values.

This approach is highly flexible, enabling valuation of European and American options, dividend-paying and

non-dividend-paying assets, and a wide range of exotic payoff structures. As the number of time steps increases, multi-period binomial model prices converge to those of continuous-time models.

4.2 Multi-Period Binomial Model Setup

Recall that the single-period binomial model assumes a single up or down move between time 0 and T , under the risk neutral measure $\mathbb{Q}$. The same can be extended by applying the same reasoning over N time steps. At each step, the stock either goes up by u or down by d , so that:

$$S_{n+1} = \begin{cases} S_n u, & \text{with probability } q_u, \\ S_n d, & \text{with probability } q_d = 1 - q_u. \end{cases}$$

under the risk-neutral measure $\mathbb{Q}$.

We will denote $S_N(\omega)$ as the stock price at time N along path (or node) ω , given by

$$S_N(\omega) = S_0 u^{N_u(\omega, N)} d^{N_d(\omega, N)}, \quad N_u + N_d = N.$$

further demonstrating the multiplicative property of the multi-period binomial model.

4.3 Risk-Neutral Recursion

Pricing a European option in a multi-period binomial model is done using a backwards-recursion method.

Let V_n^Λ denote the option value at node Λ and time n . The option price is obtained recursively by discounting the expected future values under the risk neutral measure. For $n < N$, we have

$$V_n^\Lambda = \frac{1}{1+R} [q_u V_{n+1}^{\Lambda u} + q_d V_{n+1}^{\Lambda d}],$$

where q_u and q_d are as defined previously. That is, the price at time 0 is the discounted sum of the nodes accounted for the risk-neutral probabilities (under $\mathbb{Q}$), similar to the single period model. We can see from the iterative process that we end up with the following equation for the price of the option at time 0.

$$V_0 = \frac{1}{(1+R)^N} \sum_{\omega \in \Omega} Q(\omega) V_N(\omega)$$

where

$$Q(\omega) = q_u^{N_u(\omega, N)} q_d^{N_d(\omega, N)}$$

is the probability of path ω occurring, with $N_u(\omega, N)$, $N_d(\omega, N)$ denoting the total number of "up" and "down" moves along ω respectively. Note that the size of the sample space $|\Omega| = 2^N$, even though at timestep N we will have $N+1$ different stock prices (due to the multiplicative property of the multi-period model). At time 0, the price of the option can be described by:

$$\begin{aligned} V_0 &= \frac{1}{(1+R)^N} \mathbb{E}^Q[V_N] = \frac{1}{(1+R)^N} \sum_{\omega \in \Omega} Q(\omega) V_N(\omega) \\ &= \frac{1}{(1+R)^N} \sum_{\omega \in \Omega} q_u^{N_u(\omega, N)} q_d^{N_d(\omega, N)} v_N(S_0 u^{N_u(\omega, N)} d^{N_d(\omega, N)}) \\ &= \frac{1}{(1+R)^N} \sum_{\omega \in \Omega} [q_u^{N_u(\omega, N)} q_d^{N_d(\omega, N)} \\ &\quad \times v_N(S_0 u^{N_u(\omega, N)} d^{N-N_u(\omega, N)})]. \end{aligned}$$

Note that from the above result we are able to see that the sum only depends on $N_u(\omega, N)$ and not the order in which the up/down moves occur. So we can change the summation variable to $j = N_u(\omega, N)$.

$$V_0 = \frac{1}{(1+R)^N} \sum_{j=0}^N \binom{N}{j} q_u^j q_d^{N-j} v_N(S_0 u^j d^{N-j})$$

4.4 European Call Option

Recall that the payoff of the option at time N is given by

$$V_N(S_N) = \max(S_N - K, 0)$$

The price of the call option at time 0 using the previous result is

$$\begin{aligned} C_0 &= \frac{1}{(1+R)^N} \sum_{j=0}^N \binom{N}{j} q_u^j q_d^{N-j} \max(S_0 u^j d^{N-j} - K, 0) \\ &= \frac{1}{(1+R)^N} \sum_{j=j_{\min}}^N \binom{N}{j} q_u^j q_d^{N-j} [S_0 u^j d^{N-j} - K] \end{aligned}$$

$$\text{where } j_{\min} = \{j : S_0 u^j d^{N-j} > K\}$$

The call option can be written as

$$\begin{aligned} C_0 &= \frac{S_0}{(1+R)^N} \sum_{j=j_{\min}}^N \binom{N}{j} q_u^j q_d^{N-j} u^j d^{N-j} \\ &\quad - \frac{K}{(1+R)^N} \sum_{j=j_{\min}}^N \binom{N}{j} q_u^j q_d^{N-j} \end{aligned}$$

Recall that the cumulative binomial function is

$$\Psi(n; N, q) = \sum_{j=n}^N \binom{N}{j} q^j (1-q)^{N-j}.$$

Using the cdf we can rewrite the price of the call option as

$$\begin{aligned} C_0 &= \frac{1}{(1+R)^N} \sum_{j=j_{\min}}^N \binom{N}{j} q_u^j q_d^{N-j} (S_0 u^j d^{N-j} - K) \\ &= S_0 T_1 - \frac{K}{(1+R)^N} T_2, \end{aligned}$$

where

$$\begin{aligned} T_1 &= \sum_{j=j_{\min}}^N \binom{N}{j} \left(\frac{q_u u}{1+R} \right)^j \left(\frac{q_d d}{1+R} \right)^{N-j}, \\ T_2 &= \sum_{j=j_{\min}}^N \binom{N}{j} q_u^j q_d^{N-j}. \end{aligned}$$

Now if we write

$$\bar{q}_u = \frac{q_u u}{1+R}, \quad \bar{q}_d = 1 - \bar{q}_u$$

. Therefore,

$$\begin{aligned} T_1 &= S_0 \sum_{j=j_{\min}}^N \binom{N}{j} \bar{q}_u^j (1 - \bar{q}_u)^{N-j} \\ &= S_0 \psi(j_{\min}; N, \bar{q}_u) \end{aligned}$$

Now we can see we have

$$C_0 = S_0 \psi(j_{\min}; N, \bar{q}_u) - \frac{K}{(1+R)^N} \Psi(j_{\min}; N, q_u).$$

which is known as the Cox-Ross-Rubinstein (CRR) equation.

4.5 Choice of Parameters

It is clear from their structure that the CRR binomial model and the Black-Scholes formula are closely related. We now begin the formal investigation of this convergence. We will begin by examining $(1+R)^N$ as $N \rightarrow \infty$. In order to relate the per-period interest rate R to the continuously-compounded rate r , we consider the following for the per period interest rate:

$$1+R = e^{r\Delta t}.$$

It follows that the accumulated value over N periods is

$$(1+R)^N = (e^{r\Delta t})^N = e^{rT}.$$

Hence the discrete-time discount factor satisfies

$$(1+R)^{-N} = e^{-rT},$$

exactly matching the continuous-time discount factor in the Black-Scholes model. Substituting into the CRR pricing formula yields

$$C_0 = S_0 \psi(j_{\min}; N, \bar{q}_u) - e^{-rT} K \psi(j_{\min}; N, q_u).$$

To establish convergence to the Black-Scholes formula, we can show that

$$\psi(j_{\min}; N, \bar{q}_u) \rightarrow N(d_1), \quad \psi(j_{\min}; N, q_u) \rightarrow N(d_2),$$

as $N \rightarrow \infty$. We continue the proof by choosing u and d to match the volatility of underlying asset for a small period of time Δt . Recall from Section 3.3,

$$\log\left(\frac{S_{t+\Delta t}}{S_t}\right) \sim \mathcal{N}\left(\left(r - \frac{1}{2}\sigma^2\right)\Delta t, \sigma^2\Delta t\right).$$

That is, the variance for a single period (or small change in time) Δt is $\sigma^2\Delta t$. To ensure that the discrete-time binomial model matches this volatility, we will choose the one-step "up" and "down" moves of the binomial model to be the positive and negative square roots of this variance:

$$\log u = +\sigma\sqrt{\Delta t}, \quad \log d = -\sigma\sqrt{\Delta t}.$$

So,

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}.$$

This choice guarantees that the one-period variance of the binomial model is $\sigma^2\Delta t$, matching the variance assumed by the BSM for a small Δt .

4.6 Convergence as $N \rightarrow \infty$

Now let J denote the number of upward moves. That is, $J \sim \text{Bin}(N, q_u)$

$$P(J \geq j_{\min}) = \Psi(j_{\min}; N, q_u) = \sum_{j=j_{\min}}^N \binom{N}{j} q_u^j (1 - q_u)^{N-j}.$$

The Central Limit Theorem states that, for large N , the binomial distribution converges to a standard normal distribution. In particular, if $J \sim \text{Bin}(N, q_u)$, then

$$\frac{J - Nq_u}{\sqrt{Nq_u(1 - q_u)}} \rightarrow N(0, 1).$$

Since we are considering the limit as $N \rightarrow \infty$ we can apply the CLT as follows:

$$\begin{aligned} \Psi(j_{\min}; N, q_u) &= \mathbb{P}\left(Z_N \geq \frac{j_{\min} - Nq_u}{\sqrt{Nq_u(1 - q_u)}}\right) \\ &\rightarrow N\left(-\frac{j_{\min} - Nq_u}{\sqrt{Nq_u(1 - q_u)}}\right) \end{aligned}$$

where $N(\cdot)$ denotes the standard normal cumulative distribution function. Solving for $j_{\min}$ gives:

$$\begin{aligned} S_0 u^{j_{\min}} d^{N-j_{\min}} &= K \\ \Rightarrow j_{\min} &= \frac{\log(K/S_0) - N \log d}{\log(u/d)} \\ &= \frac{N}{2} + \frac{\log(K/S_0)}{2\sigma\sqrt{\Delta t}}. \end{aligned}$$

Now we need to show that

$$\frac{j_{\min} - Nq_u}{\sqrt{Nq_u(1-q_u)}} = -d_2.$$

To evaluate the limit, we introduce Taylor series expansions. Note that the Taylor series expansion for e^x is in the form $e^x \approx 1 + x + x^2/2 + \dots$

We can expand our choices of u and d as $u = e^{\sigma\sqrt{\Delta t}}$ and $d = e^{-\sigma\sqrt{\Delta t}}$ as follows:

$$\begin{aligned} u &= 1 + \sigma\sqrt{\Delta t} + \frac{1}{2}\sigma^2\Delta t + O(\Delta t^{3/2}), \\ d &= 1 - \sigma\sqrt{\Delta t} + \frac{1}{2}\sigma^2\Delta t + O(\Delta t^{3/2}). \end{aligned}$$

and since $1 + R = e^{r\Delta t}$,

$$1 + R = 1 + r\Delta t + O(\Delta t^2).$$

Now it's easy to see that

$$\begin{aligned} u - d &= 2\sigma\sqrt{\Delta t} + O(\Delta t^{3/2}), \\ 1 + R - d &= \sigma\sqrt{\Delta t} + (r - \frac{1}{2}\sigma^2)\Delta t + O(\Delta t^{3/2}). \end{aligned}$$

Using the newly chosen terms we have:

$$\begin{aligned} \Rightarrow q_u &= \frac{1}{2} + \frac{(r - \frac{1}{2}\sigma^2)\sqrt{\Delta t}}{2\sigma} + O(\Delta t). \\ \Rightarrow \sqrt{Nq_u(1-q_u)} &= \frac{\sqrt{N}}{2} + O(N^{-1/2}). \\ \Rightarrow Nq_u &= \frac{N}{2} + \frac{(r - \frac{1}{2}\sigma^2)T}{2\sigma\sqrt{T}}\sqrt{N} + O(1). \end{aligned}$$

Note that we've already begun simplifying the terms (combining the $O(\cdot)$ terms). Recall from the central limit theorem we have

$$\frac{j_{\min} - Nq_u}{\sqrt{Nq_u(1-q_u)}} \sim N(0, 1)$$

and our goal is to show it's convergence to d_2 . Now expanding the fraction using the Taylor approximation gives

$$\begin{aligned} &\frac{j_{\min} - Nq_u}{\sqrt{Nq_u(1-q_u)}} \\ &= \frac{\left[\frac{N}{2} + \frac{\log(K/S_0)}{2\sigma\sqrt{\Delta t}} \right] - \left[\frac{N}{2} + \frac{(r - \frac{1}{2}\sigma^2)T}{2\sigma\sqrt{T}}\sqrt{N} + O(1) \right]}{\frac{\sqrt{N}}{2} + O(N^{-1/2})} \\ &= \frac{\frac{\sqrt{N}}{2\sigma\sqrt{T}} \left[\log(K/S_0) - (r - \frac{1}{2}\sigma^2)T \right] - O(1)}{\frac{\sqrt{N}}{2} + O(N^{-1/2})}. \end{aligned}$$

We drop the $O(\cdot)$ terms to continue the proof. Note that we

are not dropping $O(1)$ directly, rather $\frac{O(1)}{\sqrt{N/2 + O(N^{-1/2})}}$. Clearly as $N \rightarrow \infty$ the fraction tends to 0. Dropping the $O(\cdot)$ terms gives:

$$\begin{aligned} &= \frac{\frac{1}{2\sigma\sqrt{T}} \left[\log(K/S_0) - (r - \frac{1}{2}\sigma^2)T \right]}{\frac{1}{2}} \\ &= \frac{\log(K/S_0) - (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \\ &= -\frac{\log(S_0/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} = -d_2. \end{aligned}$$

This establishes

$$\Psi(j_{\min}; N, q_u) \rightarrow N(d_2).$$

as required.

4.7 Convergence of $\Psi(j_{\min}; N, \bar{q}_u)$ to $N(d_1)$.

The same reasoning works to show

$$\Psi(j_{\min}; N, \bar{q}_u) \rightarrow N(d_1).$$

By the central limit theorem we can show that

$$\frac{j_{\min} - N\bar{q}_u}{\sqrt{N\bar{q}_u(1-\bar{q}_u)}} \rightarrow N(0, 1).$$

Now using Taylor expansions as derived previously, we can show that:

$$\bar{q}_u = \frac{1}{2} + \frac{(r + \frac{1}{2}\sigma^2)}{2\sigma}\sqrt{\Delta t} + O(\Delta t)$$

Now we can see that,

$$\begin{aligned} \Rightarrow N\bar{q}_u &= N \times \left[\frac{1}{2} + \frac{(r + \frac{1}{2}\sigma^2)}{2\sigma}\sqrt{\Delta t} + O(\Delta t) \right] \\ &= \frac{N}{2} + \frac{(r + \frac{1}{2}\sigma^2)T}{2\sigma\sqrt{T}}\sqrt{N} + O(1) \\ \Rightarrow \sqrt{N\bar{q}_u(1-\bar{q}_u)} &= \frac{\sqrt{N}}{2} + O\left(\frac{1}{\sqrt{N}}\right) \end{aligned}$$

as $N \rightarrow \infty$. Now using the CLT, and the above result we have

$$\begin{aligned} &\frac{j_{\min} - N\bar{q}_u}{\sqrt{N\bar{q}_u(1-\bar{q}_u)}} = -d_1 \\ &= \frac{\left[\frac{N}{2} + \frac{\log(K/S_0)}{2\sigma\sqrt{\Delta t}} \right] - \left[\frac{N}{2} + \frac{(r + \frac{1}{2}\sigma^2)T}{2\sigma\sqrt{T}}\sqrt{N} + O(1) \right]}{\frac{\sqrt{N}}{2} + O(N^{-1/2})} \\ &= \frac{\frac{\sqrt{N}}{2\sigma\sqrt{T}} \left[\log(K/S_0) - (r + \frac{1}{2}\sigma^2)T \right] - O(1)}{\sqrt{N}/2 - O(N^{-1/2})} \end{aligned}$$

Dropping the $O(\cdot)$ terms, which vanish as $N \rightarrow \infty$ by the same argument as before, we obtain

$$\begin{aligned} &= \frac{\log(K/S_0) - (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \\ &= -\left[\frac{\log(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \right] = -d_1 \end{aligned}$$

as required. Thus as $N \rightarrow \infty$,

$$\psi(j_{\min}, N, \bar{q}_u) \rightarrow N(d_1).$$

4.8 Concluding the proof

Now we've derived the necessary results to show that the Multi-Period Binomial Model converges to BSM for European call option. Recall the CRR formula,

$$C_0 = S_0 \Psi(j_{\min}; N, \bar{q}_u) - \frac{K}{(1+R)^N} \Psi(j_{\min}; N, q_u),$$

We've shown that as $N \rightarrow \infty$

$$\Psi(j_{\min}; N, \bar{q}_u) \rightarrow N(d_1)$$

and

$$\Psi(j_{\min}; N, q_u) \rightarrow N(d_2)$$

Noting that $(1+R)^N \rightarrow e^{rT}$ as $N \rightarrow \infty$, we see that the CRR converges to Black-Scholes. That is,

$$C_0 = S_0 \Psi(j_{\min}; N, \bar{q}_u) - \frac{K}{(1+R)^N} \Psi(j_{\min}; N, q_u)$$

$$\rightarrow C_0 = S_0 N(d_1) - K e^{-rT} N(d_2).$$

4.9 Extension to the European Put

We can use the results proven during the proof of convergence for the European Call to show the same convergence holds for the European put option. Rather than repeating the full argument, we can see from the put-call parity, that considering a multiperiod model with N timesteps we have:

$$C_0^{(N)} - P_0^{(N)} = S_0 - \frac{K}{(1+R)^N}.$$

As $N \rightarrow \infty$, both the stock and the discount factor converge to their continuous-time limits, so

$$C_0 - P_0 = S_0 - K e^{-rT}.$$

Since we have already shown that $C_0^{(N)} \rightarrow C_0$, it directly follows that

$$P_0^{(N)} \rightarrow P_0 = K e^{-rT} N(-d_2) - S_0 N(-d_1),$$

where d_1 and d_2 are the same as in the call formula. Hence, the binomial model also converges to the Black-Scholes value for the European put.

5 C++ Implementation

5.1 Objective of the program

The code follows the option pricing concepts as discussed throughout the report. We intend to show that the Cox-Ross-Rubinstein expression derived from the multi-period binomial model converges to the Black-Scholes model as we increase the number of steps (N) taken to reach the delivery time.

5.2 Files

main.cpp file consists of the function definitions that calculate the call and put option prices using the Black-Scholes model, the N -period Binomial Model and finally the functions to be used by the said pricing models.

OptionParameters.h file contains the parameter definitions stock price (S), strike price (K), time to Expiry (T), risk free rate (r) and the volatility (σ). This class would be the basic input for both the models, while the binomial model takes in another input parameter, the number of steps.

plots.ipynb is the python notebook that we used to plot all our data that we produced in the c++ project. We have also made plots to visualise the interesting natures of hyperparameters (S, K, T, r, σ) affecting the convergence of binomial model to the Black-Scholes model.

5.3 Data Plotting and Inferences

5.3.1 The General trend of Models

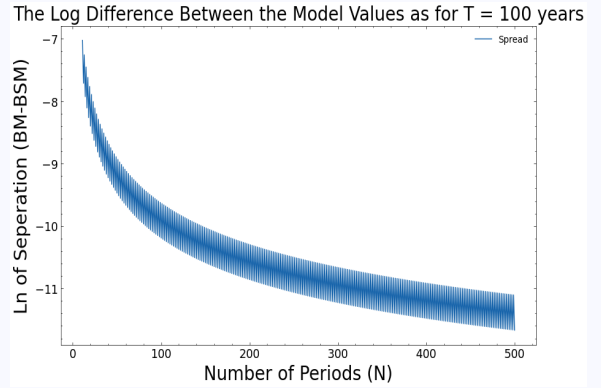


Figure 1: Depicts the general trend of convergence of the models for a Put Option, measured by the logarithm of the difference in prices.

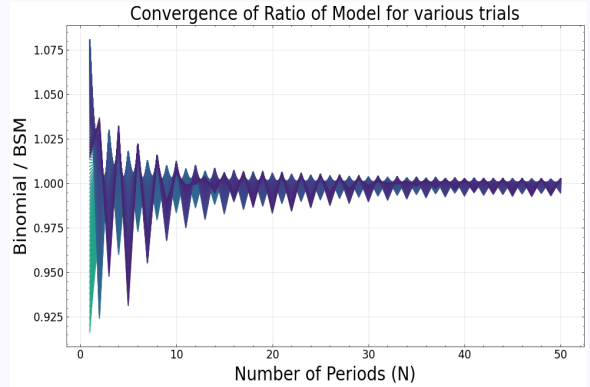


Figure 2: Depicts the general trend of convergence of the models for a Call Option, in the initial steps ($N < 50$).

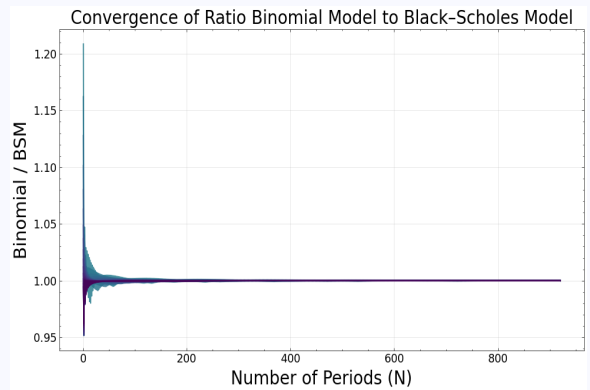


Figure 3: Depicts the general trend of convergence of the models for a Call Option for a larger range of N .

Figure 1 shows the convergences of the natural log of the difference between the option prices of Binomial model and the Black-Scholes model ($\rightarrow 0$), for put options. Figures 2 and 3 show the general trend of convergence of the N -period binomial model towards the Black-Scholes model ($\rightarrow 1$), for call options. The Expression we have used to evaluate them is N -period Binomial price divided by the Black-Scholes price.

Inferences: We can notice the spiky plots as N value increases. This is because of the nature of the Binomial

model, the centre oscillates around the strike price as the number of steps increases. Our Literature review pointed out that this improves if we use a certain type of trinomial model, which has better centring and lesser oscillations around the strike price. It is also good to know that the trinomial model converges faster than the binomial model.

5.3.2 Other Observations around the parameters

We tried looking at how each parameter affects the convergence of the Binomial model to Black-Scholes model for Call options mainly in this report. We observed some interesting scenarios that can also be proved by working around the Cox-Ross-Rubenstein Expression.

- Options with a longer expiries (high T), low stock prices relative to the strike price, or low volatility converge from the bottom, that is, the Binomial model price is mostly lesser than the Black-Scholes model price as we iterate.
- The prices of Options that are further out of the money are slower to converge than those that are closer to or in-the-money. The separation continues to be larger until $N \approx 300$.
- Convergence of options with lower volatility seems the strictest (<1), as the Binomial Option price moves up, closer to the Black-Scholes price (Figure 5). this might be because of the per period move sizes (u, d) decreasing as it depends on the σ value.

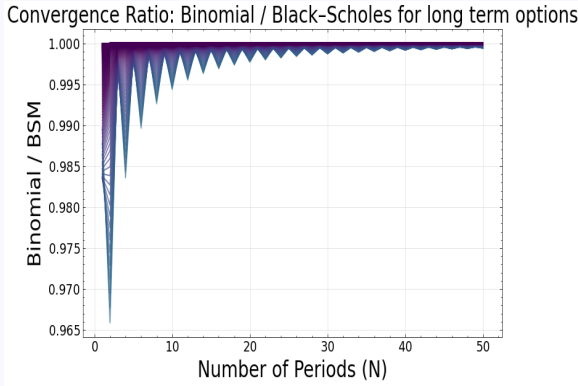


Figure 4: Convergence for trials where $T > 15$ years for a Call Option.

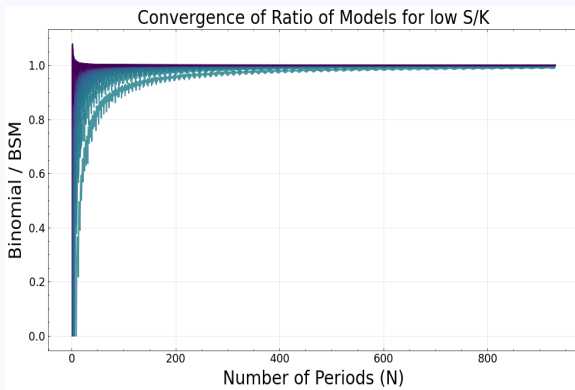


Figure 5: Convergence for low Stock prices relative to Strike prices for a Call Option.

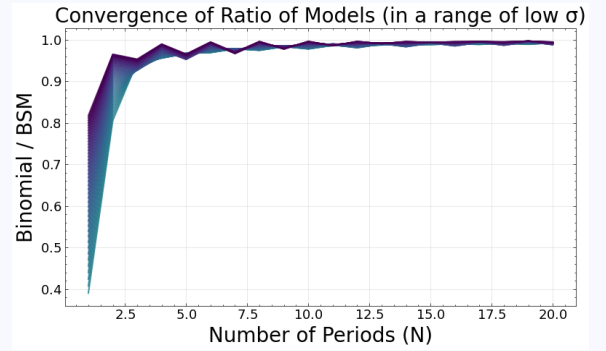


Figure 6: Convergence for low volatility (σ) conditions for a Call Option.

6 American Options and Extensions of the Models

As mentioned in section 1, the key difference between American and European Options is that American options are able to be exercised prior to expiry, whereas European Options can only be exercised at expiry. This property makes American options inherently harder to price/value.

6.1 Extension to the Binomial Model

6.1.1 Single-Period Binomial Model

It is easy to see that under the single-period binomial pricing model, the American Option will behave the same as the European option - that is, it will be exercised if it is in the money at time T , or it will not be exercised if it is out of the money at time T .

6.1.2 Pricing American Options under the Multi-Period Binomial Model

Recall the setup for the multi-period binomial model and the corresponding recursive algorithm to price European options 4.3. Pricing American options requires a similar backward-recursive method, however at each node we must consider the owner's right to exercise early. At each node we assume the holder acts rationally, that is, the holder will make the choice that maximizes the option's value.

At expiry, the value at each terminal node is simply the intrinsic value $g(S_{N,i})$. For earlier times, we must compare the immediate exercise value with the continuation value (the discounted expected value under the risk-neutral measure). Thus, at each node (n, i) we set

$$V_{n,i} = \max \left(g(S_{n,i}), \frac{1}{1+R} (q_u V_{n+1,i+1} + q_d V_{n+1,i}) \right)$$

$$n = N-1, \dots, 0.$$

This recursion accounts for the owner's ability to rationally exercise the option at any time before maturity. In order to price the option at $t = 0$ we must calculate the values along the nodes until we have $V_{1,0}$ and $V_{1,1}$, then under the risk neutral measure $\mathbb{Q}$ we can see that

$$V_0 = \frac{1}{1+R} \left[q_u V_{1,1} + q_d V_{1,0} \right]$$

The flexibility of the multi-period binomial model allows the additional components involved with pricing American options to be incorporated seamlessly.

6.2 Extension to the Black-Scholes Model

6.2.1 American Calls

Consider an American option whose underlying does not pay any dividends. In this case, it is never optimal to exercise the American call option early, so the Black-Scholes model is appropriate for pricing such an option.

Now consider an American call option whose underlying asset pays dividends. In this case, there may be circumstances in which early exercise becomes optimal. Fisher Black proposed an approximation to handle this situation, providing a practical way to estimate the value of an American option when dividends affect the exercise decision.

6.2.2 American Puts

However, for the American put option there are circumstances in which acting rationally would mean exercising the option early. For that reason, the BSM model is not applicable to accurately price American put options, so the recursive binomial method remains the appropriate valuation approach. In practice, market participants choose to use the Black-Scholes price as a lower bound for the value of an American put.

6.3 Conclusion & Further Extensions

In practice, the binomial tree model is preferred for American options due to its flexibility, while the Black-Scholes model is used for analytical insight and as a reference benchmark. Combining both methods allows effective pricing, hedging, and risk management for American options under realistic market conditions.

The binomial tree model discretizes possible asset price paths, Black-Scholes provides a closed-form formula for European options, and Monte Carlo simulation uses risk-neutral valuation to simulate a large number of paths for the underlying asset price to price options for which no closed-form solution is available, such as American or exotic options.

On the other hand, the binomial model can lead to large errors because it is limited by two states. As mentioned in section 5.3.1, the Trinomial pricing model was introduced to help counter these larger errors (Boyle, 1977). The trinomial model was modified by Boyle and Lau who proposed that it is more accurate to consider three future (next step) price states (Boyle and Lau, 1977). Their argument suggests that the trinomial model converges to the correct value faster and smoother than the binomial model for pricing options.

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